Exercise 1. [Count of bissextile years] A year is bissextile if it is divisible by 4 and not by 100 unless it is divisible by 400. Develop a Python function:

- countBissextile(year1, year2) that returns the number of bissextile years in the range [year1, year2).

Exercise 2. Consider a list L of triplets (e, s, v) where e is a string with the name of a university course, s is a student-id, and v is the grade of the student who passed the exam of the course. Write a Python function summary(L) that returns a dictionary mapping:

- the pair ("avg", e) to the average of grades for course e
- the pair ("top", e) to the list of students that passed exam e with top score.

For example:

summary([('pds', 1, 28), ('pds', 2, 25), ('pds', 3, 28), ('dm', 1, 22), ('dm', 3, 30)]) =
{ ('avg, 'pds'): 27, ('avg, 'dm): 26, ('top, 'pds'): [1,3], ('top', 'dm')=[3] }

Exercise 3. [grouping] Consider a list of dictionaries data . Each dictionary maps the course a student has passed to the grade obtained (>= 18). E.g.,

```
[
    { 'pds':20, 'dm': 30},
    {'log':28, 'pds':30},
    ...
]
```

Develop a Python function:

- gby(data) that returns a dictionary mapping every course to the average grade of students who passed that course.

Exercise 4. Develop a Python function nequals1(l) that, given a list l, returns a list r such that r[i] is the number of elements in l that are equal to l[i].

For example, nequals1(['a', 'b', 'a', 'c', 'a', 'b']) = [3, 2, 3, 1, 3, 2]

nequals1(l) must not use list comprehension. Then develop an equivalent function nequals2(l) which must use list comprehension.